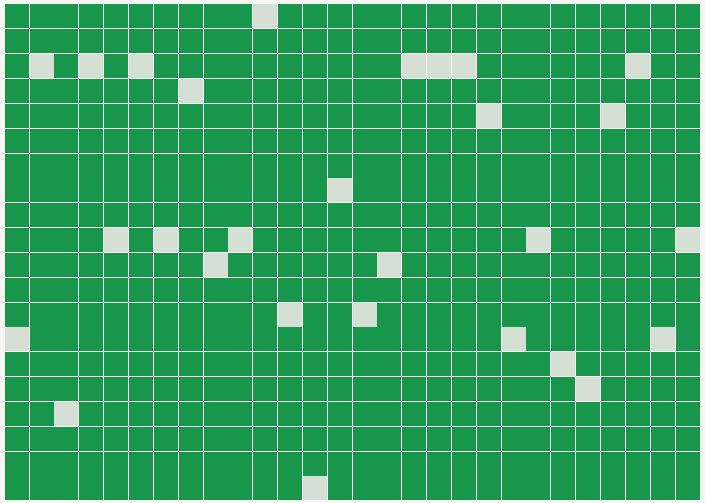
CT553H · RESEARCH EXPLAINED

# RESIDUE-LEVEL DEGRON MUTATION EFFECT PREDICTION AND ANALYSIS TOOL

Machine learning to predict ΔPSI
A tool for prioritizing experiments

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Supervisor: Dr. Hồ Quang Thái
Can Tho University · 3 Nov 2026



One cell · one substitution

BIOLOGY AND THE PREDICTION TASK

# A protein is built from an amino acid sequence



Protein structure model · illustration

## Protein → peptide → residue

| M | A | G | L | S | K | V | T |
|---|---|---|---|---|---|---|---|

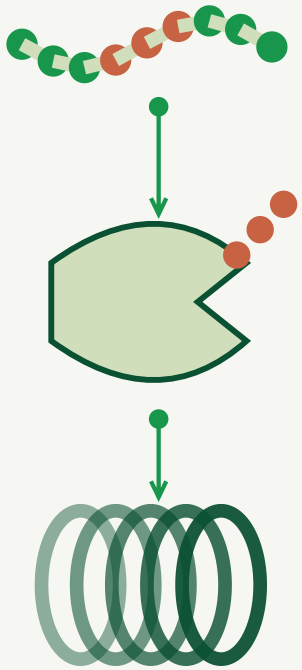
One letter · one amino acid at one position

A peptide is a short amino acid chain.
A residue is one amino acid unit in the chain.

The model reads a sequence of letters, each representing an amino acid.

BIOLOGY AND THE PREDICTION TASK

# Degrons help direct protein degradation



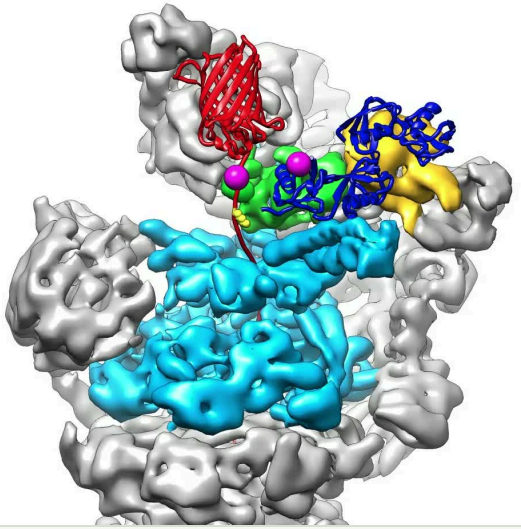
**Protein with a degron**

**Recognition and ubiquitination**

By the appropriate machinery

**Proteasome**

Cleavage into short peptides



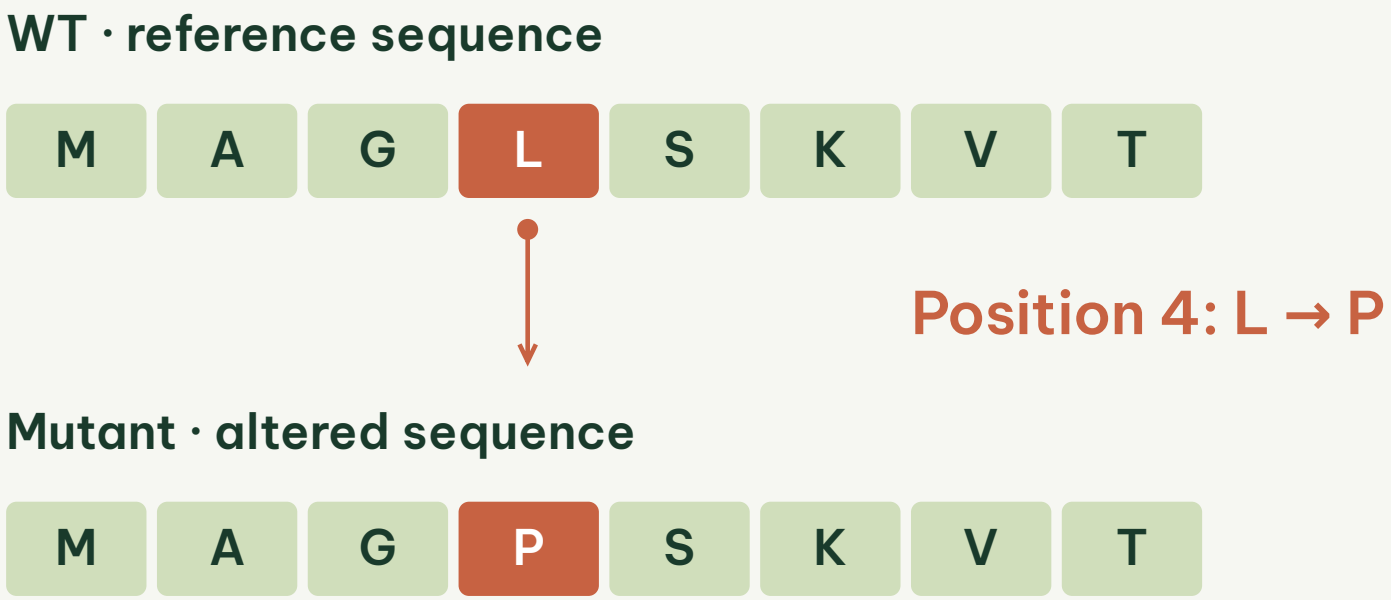
Andreas Martin / HHMI · NIGMS · molecular simulation

A degron is a signal that helps the appropriate machinery recognize a degradation substrate.

---

Recognition → ubiquitination → processing by the proteasome.

# One substitution can change the signal




**One substitution**

position
reference AA
replacement AA

WT is the reference; the mutant differs at one amino acid.

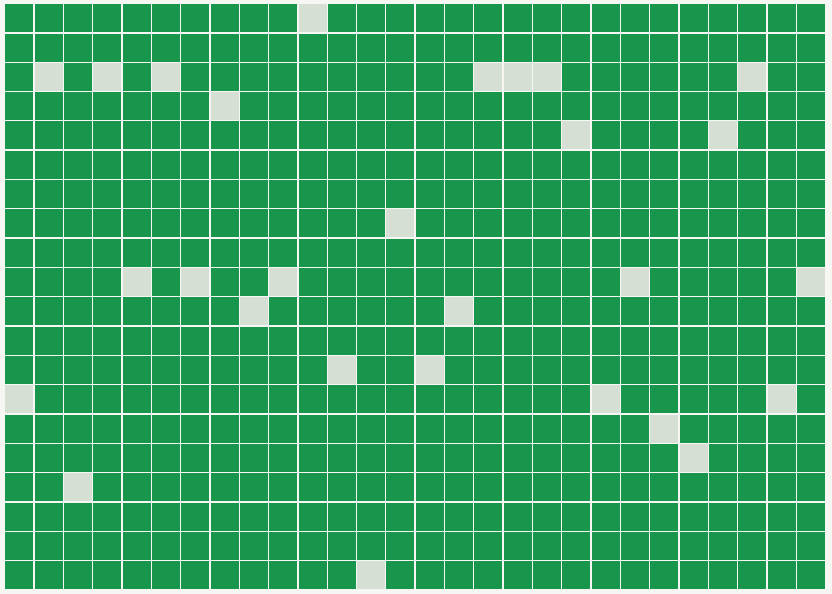
BIOLOGY AND THE PREDICTION TASK

# 532 substitutions motivate experimental prioritization

**28 positions × 19 non-WT AAs**

**532**

possible single substitutions



**Each cell is one substitution**

Segment activity alone does not identify which substitution to test first.

BIOLOGY AND EXPERIMENTAL LABELS

# GPS converts fluorescence signals into PSI

Assay schematic · the histogram is illustrative

## A shared reporter system




Reference

Signal reflects reporter stability

## GFP / DsRed ratio




Six bins · from lower to higher stability

Normalized read distribution → PSI ∈ [1; 6]

Higher PSI means a more stable reporter in the Zhang assay.

BIOLOGY AND EXPERIMENTAL LABELS

# ΔPSI is the effect the model aims to predict

Hypothetical example

Mutant **3.8** − WT **2.5**

**ΔPSI = 3.8 − 2.5 = +1.3**

Positive: the mutant reporter is more stable in this assay

---

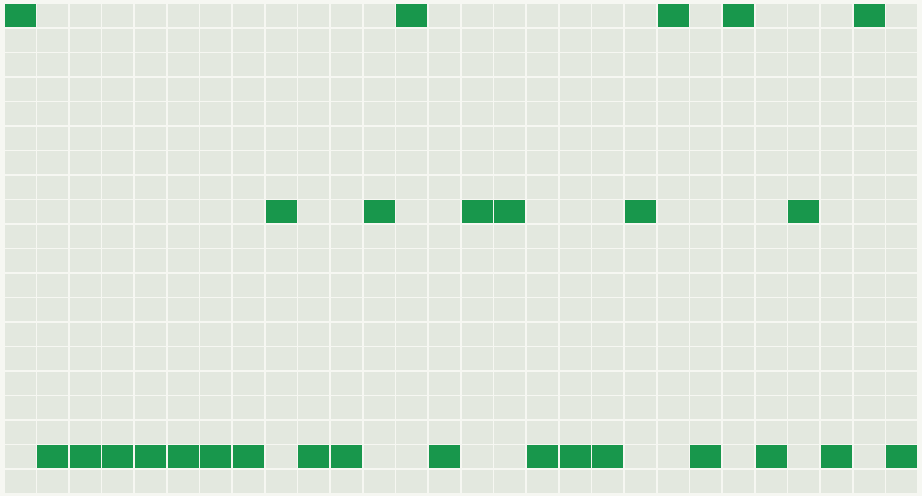
The output is a continuous value: a regression task.

DATA

# Saturation and scanning provide complementary coverage

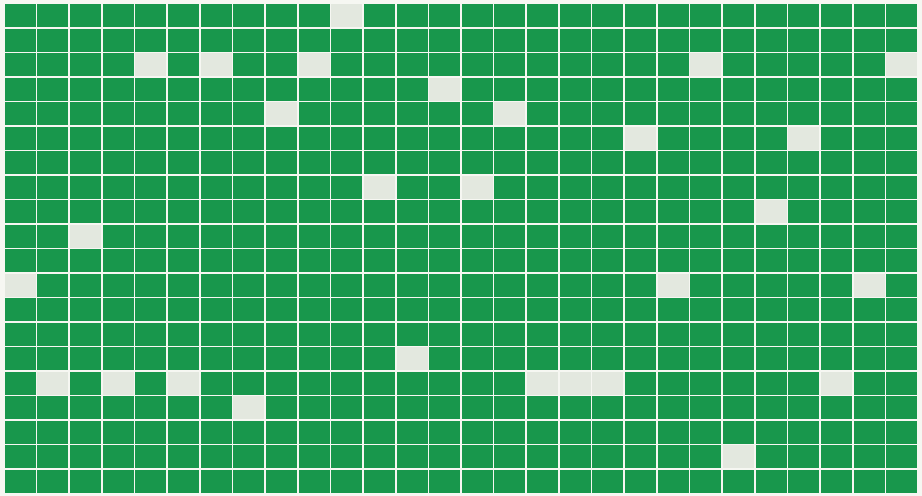
Design schematic · colored cells indicate assayed substitutions

## Scanning



One substitution per position
20 WT → mutant types

## Saturation



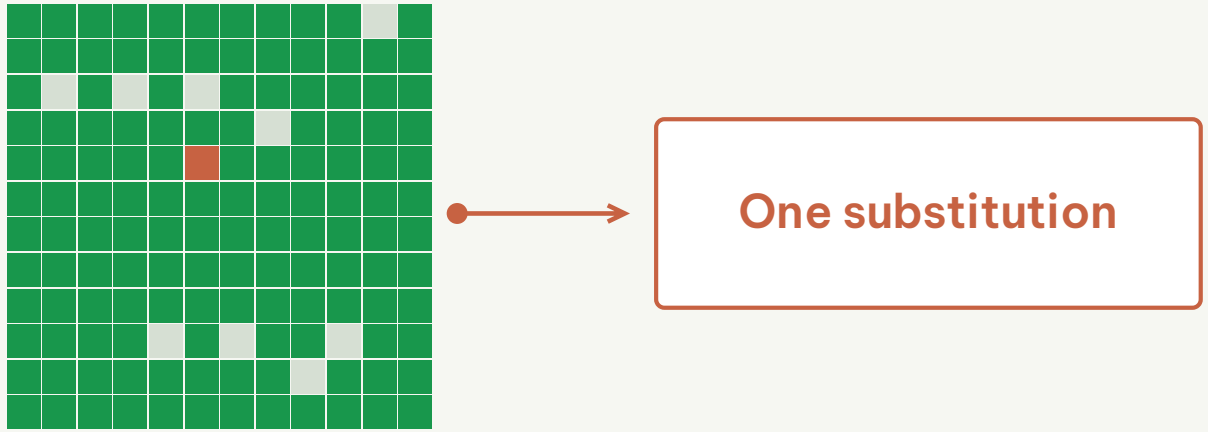
Many substitutions per position
380 WT → mutant types

Saturation covers diverse replacements; scanning spans more sequence contexts.

DATA

# Each mutation cell becomes a labeled data record

Example record key · illustrative sequence values




## Sample identity

| | |
|---|---|
| **Peptide** | one sequence context |
| **Position** | 4 |
| **Substitution** | L → P |
| **Label** | PSI(mutant) − PSI(WT) |

Match the peptide, position, substitution and WT reference before calculating ΔPSI.

DATA

# The QC-filtered data retain the full PSI range

## SATURATION

**122,412**

mutation cells · **235 peptides**

Benchmark for ΔPSI prediction

## SCANNING

**224,470**

mutation cells after QC

Transfer learning and held-source evaluation

The 235 saturation peptides form **221 protein groups** in W.
Keep technical QC; use PSI thresholds to study training scope.

Mutation cells, peptides and protein groups are different counting units.

MACHINE LEARNING WORKFLOW

# The pipeline connects data, models and the analysis tool

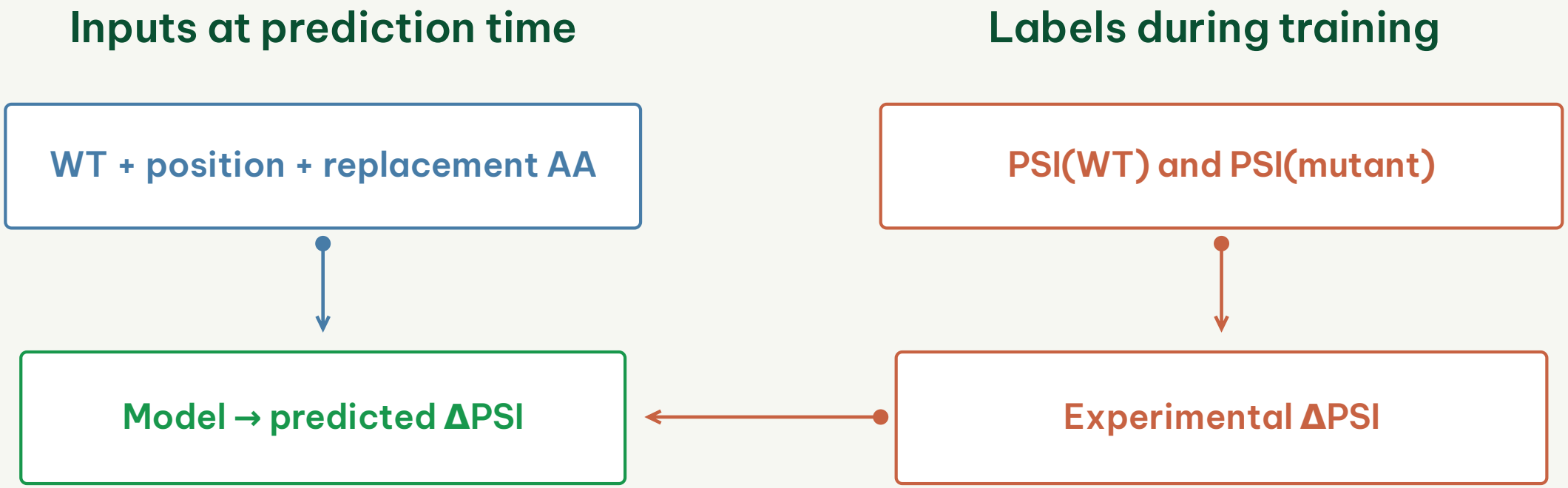



Group-based evaluation → Lock and validate → Analysis tool

Each processing step needs a rationale, an output and traceable evidence.

MACHINE LEARNING WORKFLOW

# Sequences are inputs; PSI measurements define the labels

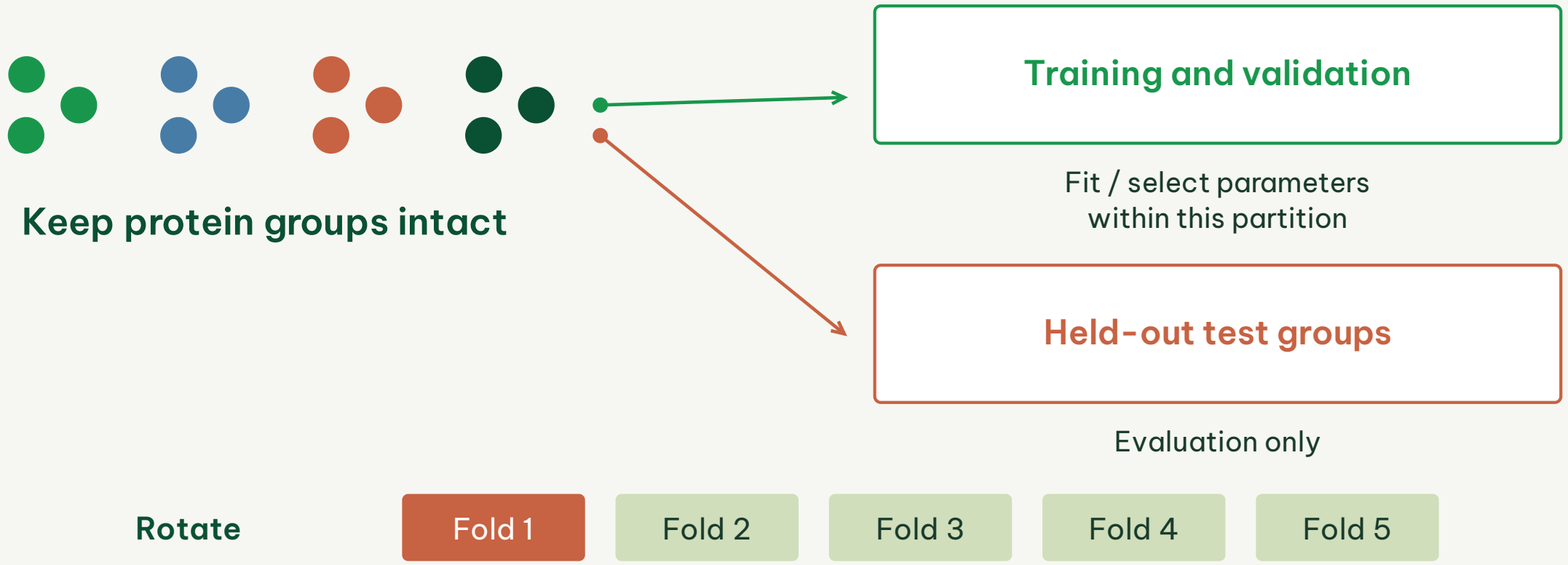



**Labels guide learning; measured PSI is not an input feature**

The model receives no experimental PSI when predicting a new substitution.

MACHINE LEARNING WORKFLOW

# Keep protein groups intact when splitting the data




Five outer folds evaluate protein groups held out from training.

MODELS AND HYPOTHESES

# Ridge tests the predictive value of sequence context




L → P

**Substitution**

WT and mutant one-hot

**Composition**

Frequencies of 20 AAs

**Context ±5**

Ordered nearby AAs

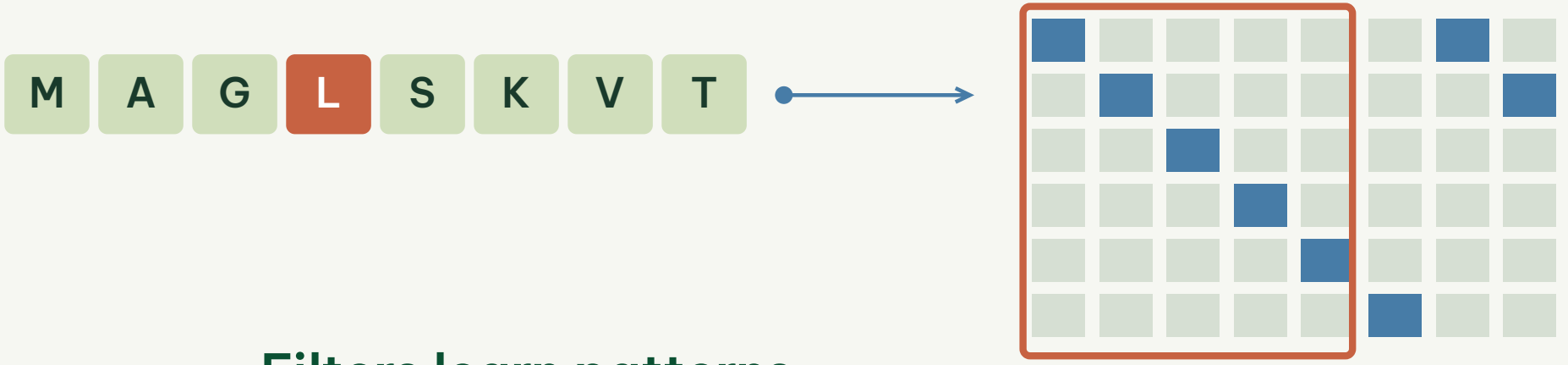
**Features × weights → one ΔPSI value**

Question: how much can sequence features and a linear model already predict?

MODELS AND HYPOTHESES

# A one-hot CNN learns local sequence patterns

Simplified one-hot schematic · not the full numeric matrix




**Filters learn patterns in local neighborhoods**

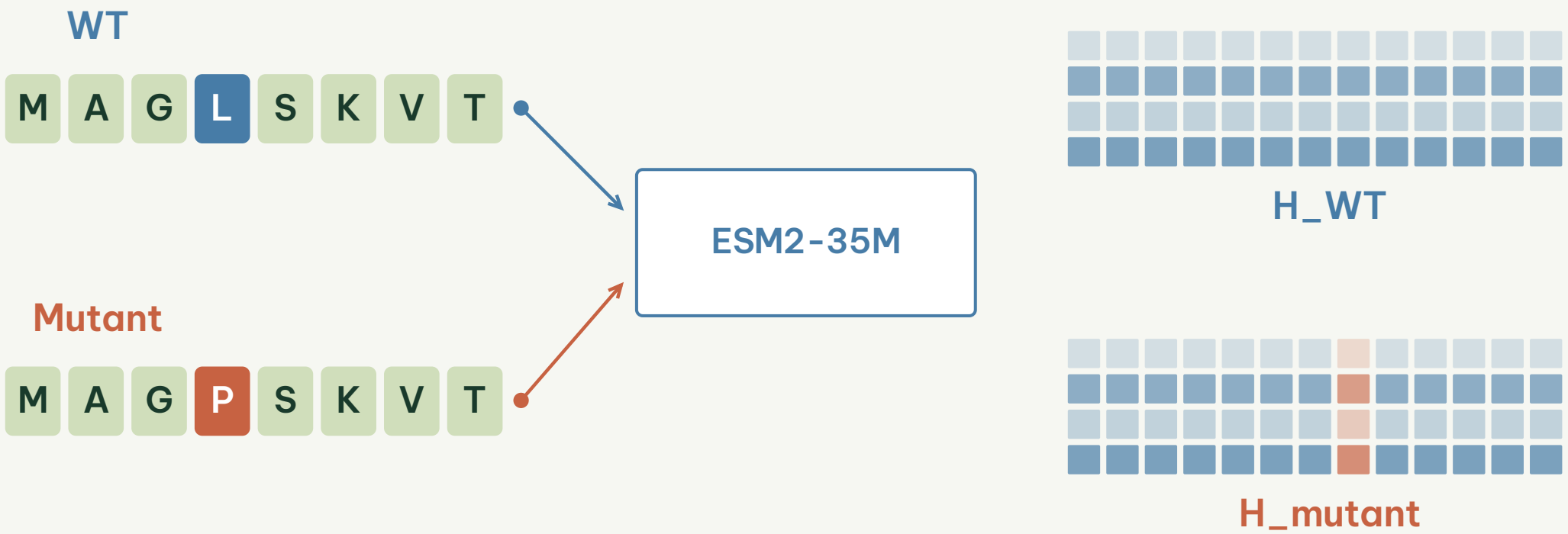
**One-hot → CNN → features → ΔPSI**

The one-hot control tests the added value of ESM representations.

MODELS AND HYPOTHESES

# ESM2 converts sequences into numeric representations

Representation schematic · colored cells illustrate vectors

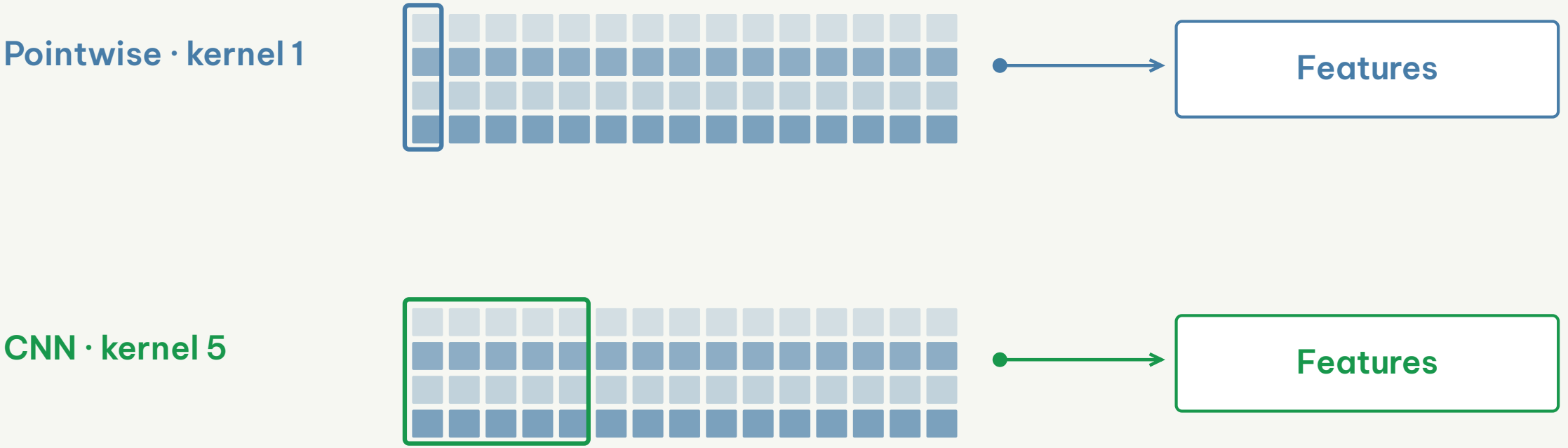



$$\Delta H = H_mutant - H_WT$$

Embedding differences describe representation changes between mutant and WT.

MODELS AND HYPOTHESES

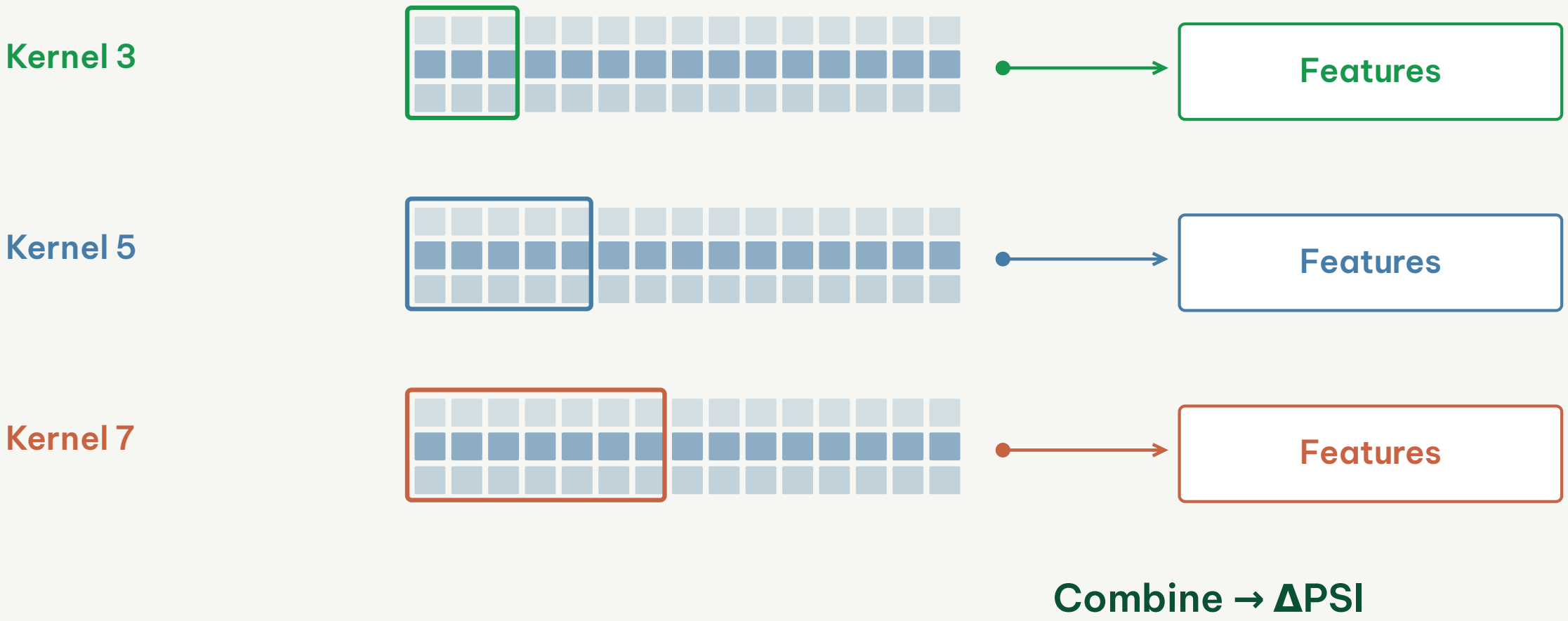
# Pointwise and CNN heads differ in position mixing




ESM embeddings already contain context; kernel 5 adds position mixing in the head.

MODELS AND HYPOTHESES

# mCNN explores multiple context widths




Question: do kernels 3, 5 and 7 improve prediction over a single kernel?

MODELS AND HYPOTHESES

# Frozen and LoRA update different parameters



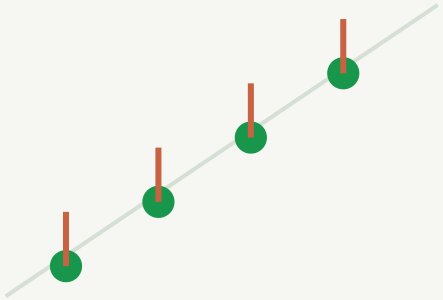

**Match seeds, folds and protocol · assess error and cost**

Assess PLM adaptation together with computational cost and stability.

EVALUATION

# Error, ranking and confidence intervals answer different questions

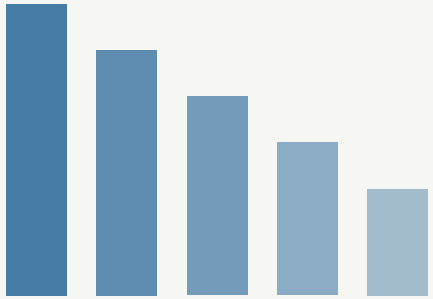
## Error



### MAE

Prediction–measurement error

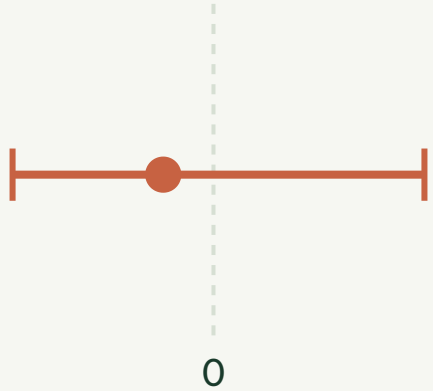
## Ranking



### AUROC / AP

Rank cells with ΔPSI ≥ 1

## Uncertainty



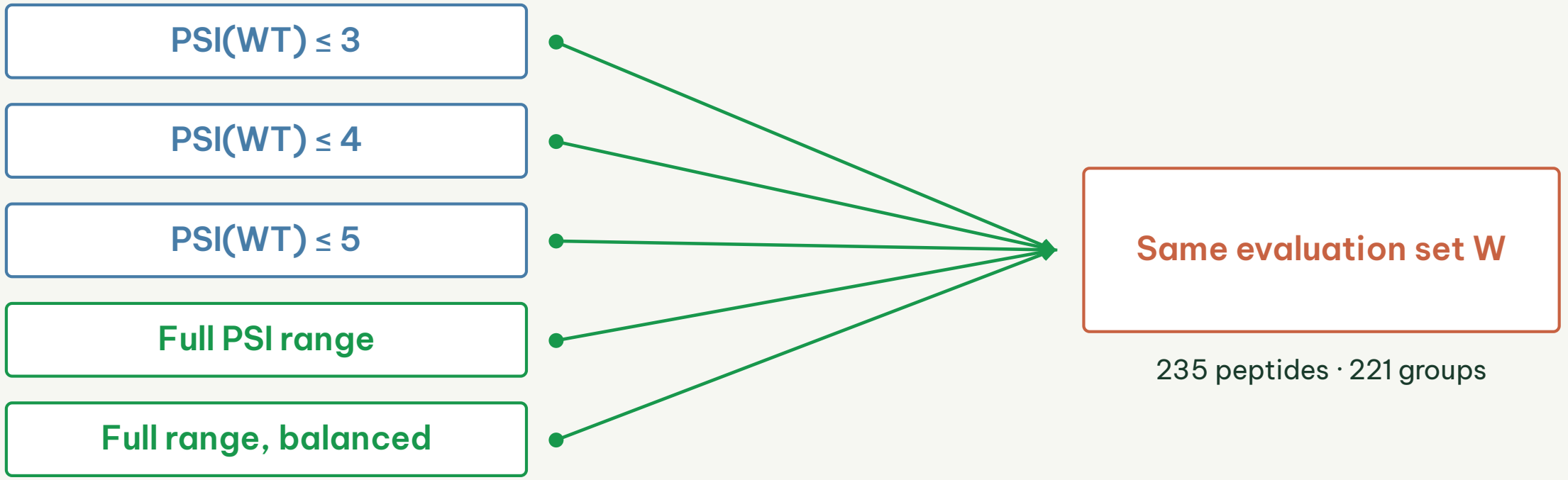

### Confidence interval

Bootstrap by protein group

MAE measures regression error · AUROC/AP assess a supplementary positive event

Lower MAE is better; higher AUROC/AP is better on eligible cohorts.

# Experiment A: which PSI range should be used for training?

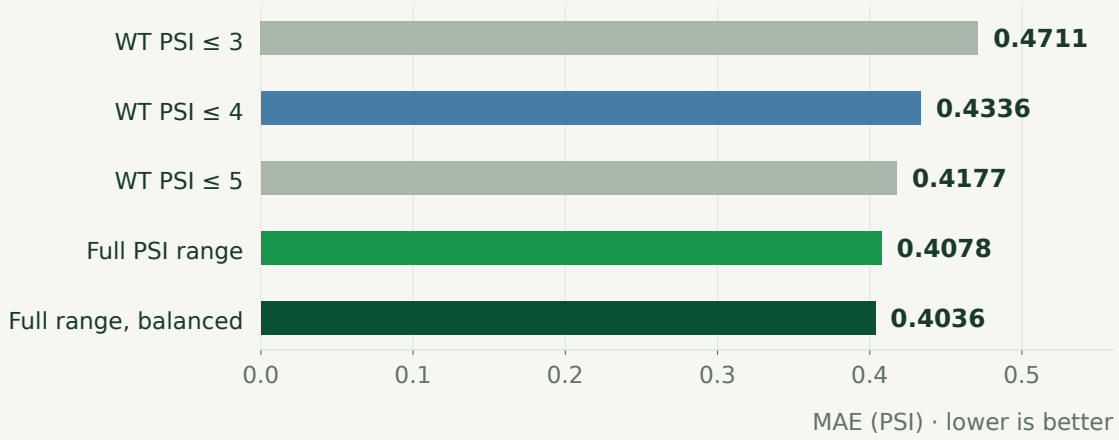



PSI thresholds define training scope; all configurations are compared on W.

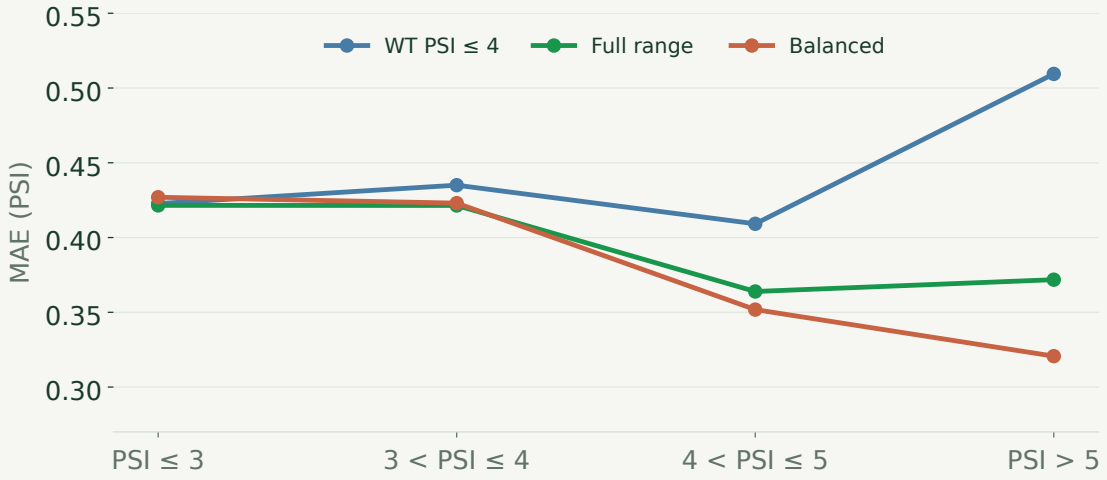
EXPERIMENTS AND RESULTS

# Full-range training reduces Ridge regression error

### Same evaluation set W · MAE ↓



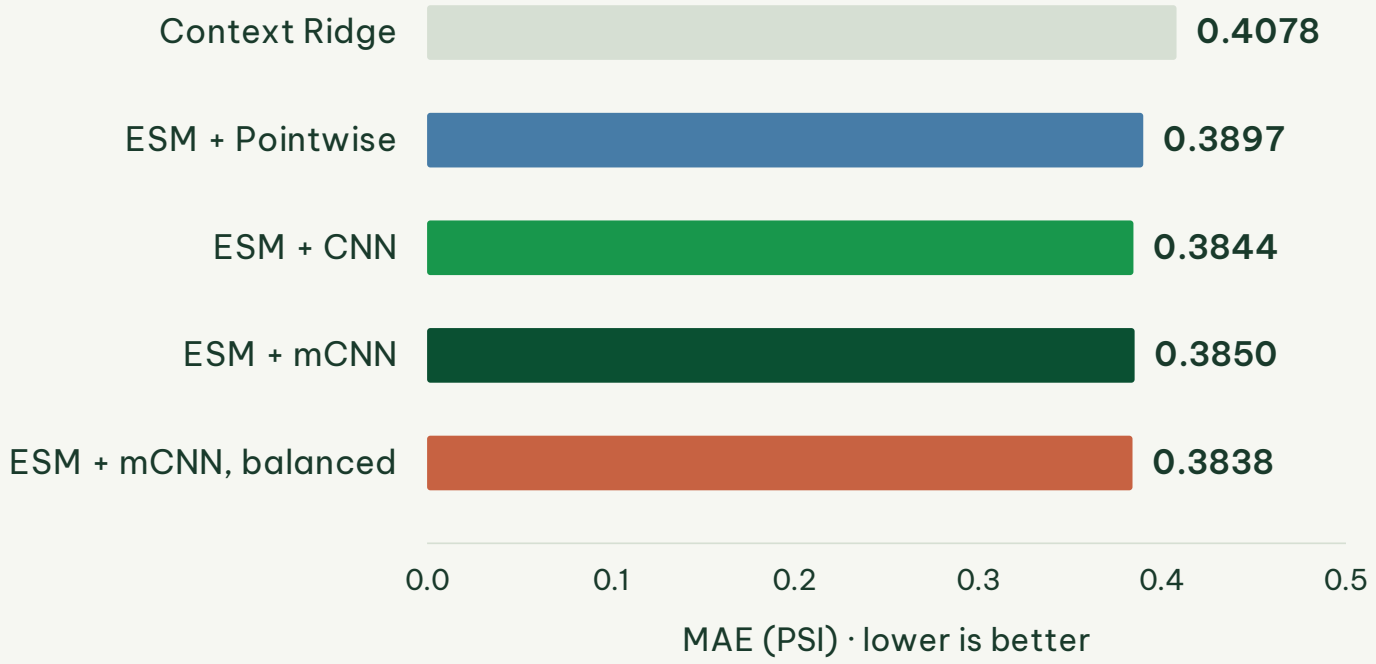

### Benefits vary across test PSI strata




Full-range training reduces MAE by about **5.94%** relative to training only on PSI ≤ 4.

EXPERIMENTS AND RESULTS

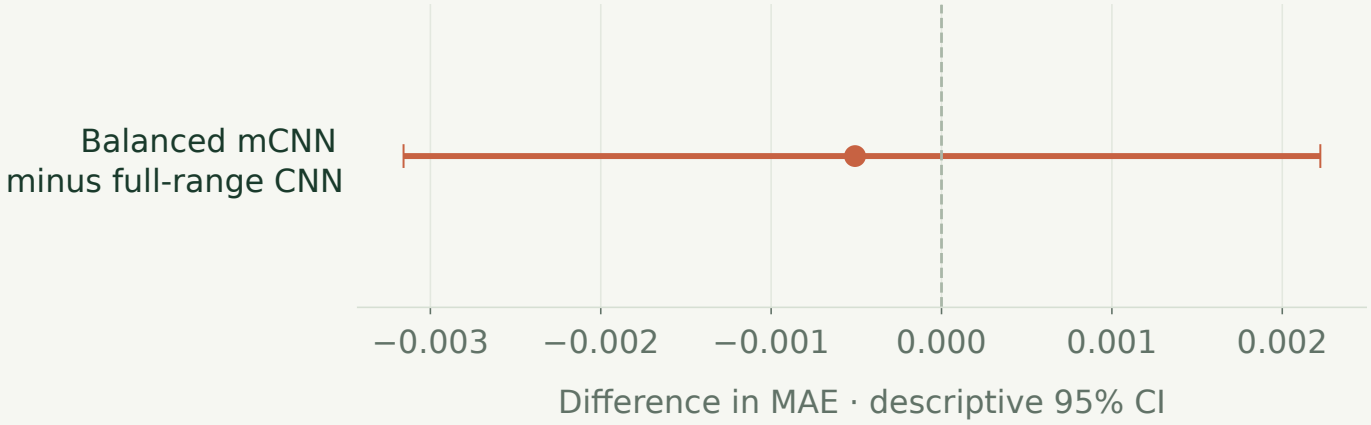
# Experiment B: what do ESM2 and CNN add?




Same W · three-seed ensemble · ESM2-35M · development results.

EXPERIMENTS AND RESULTS

# Small differences need confidence intervals




**−0.000508**

PSI · difference in MAE

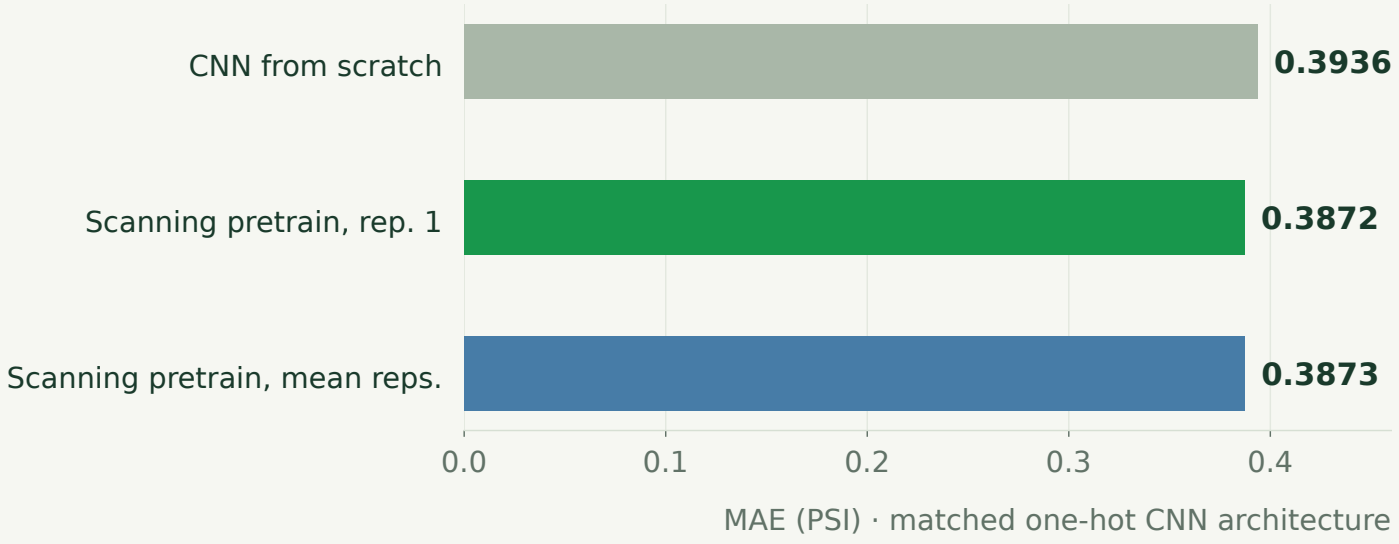
The CI crosses **0**.

Training weights also differ.

The evidence does not establish that mCNN outperforms CNN.

EXPERIMENTS AND RESULTS

# Experiment C: open questions about transfer learning




**Scanning → saturation**
Improves on one-hot CNN training from scratch.

**ESM + scanning**
The combined approach was not evaluated in this snapshot.

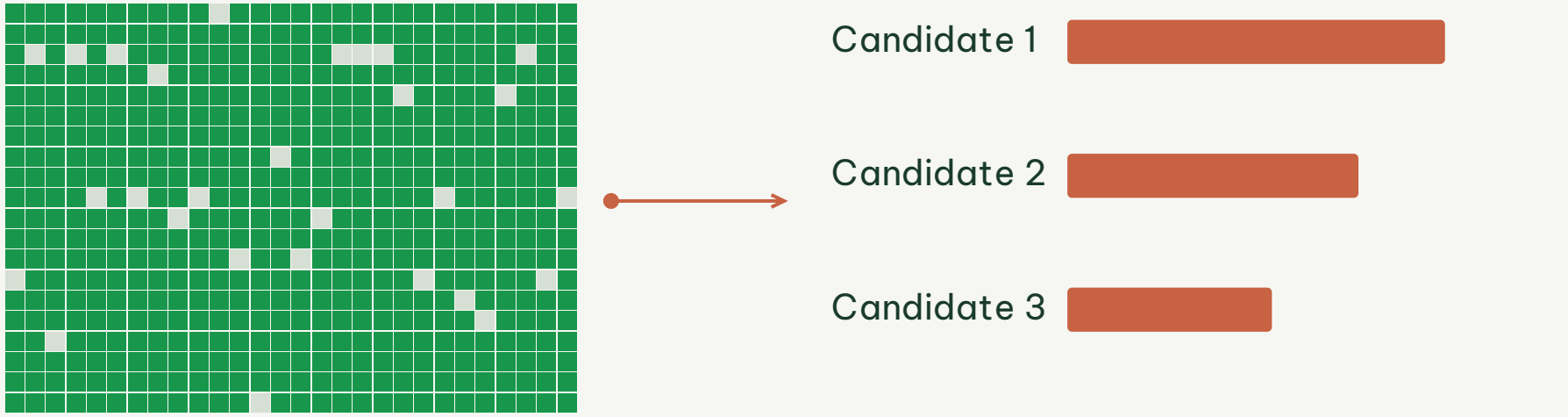
**LoRA versus frozen**
OOF results are incomplete.

Transfer benefits depend on inputs, design and matched controls.

ANALYSIS TOOL

# The effect map supports experimental prioritization

Illustrative analysis workflow




**Effect map**

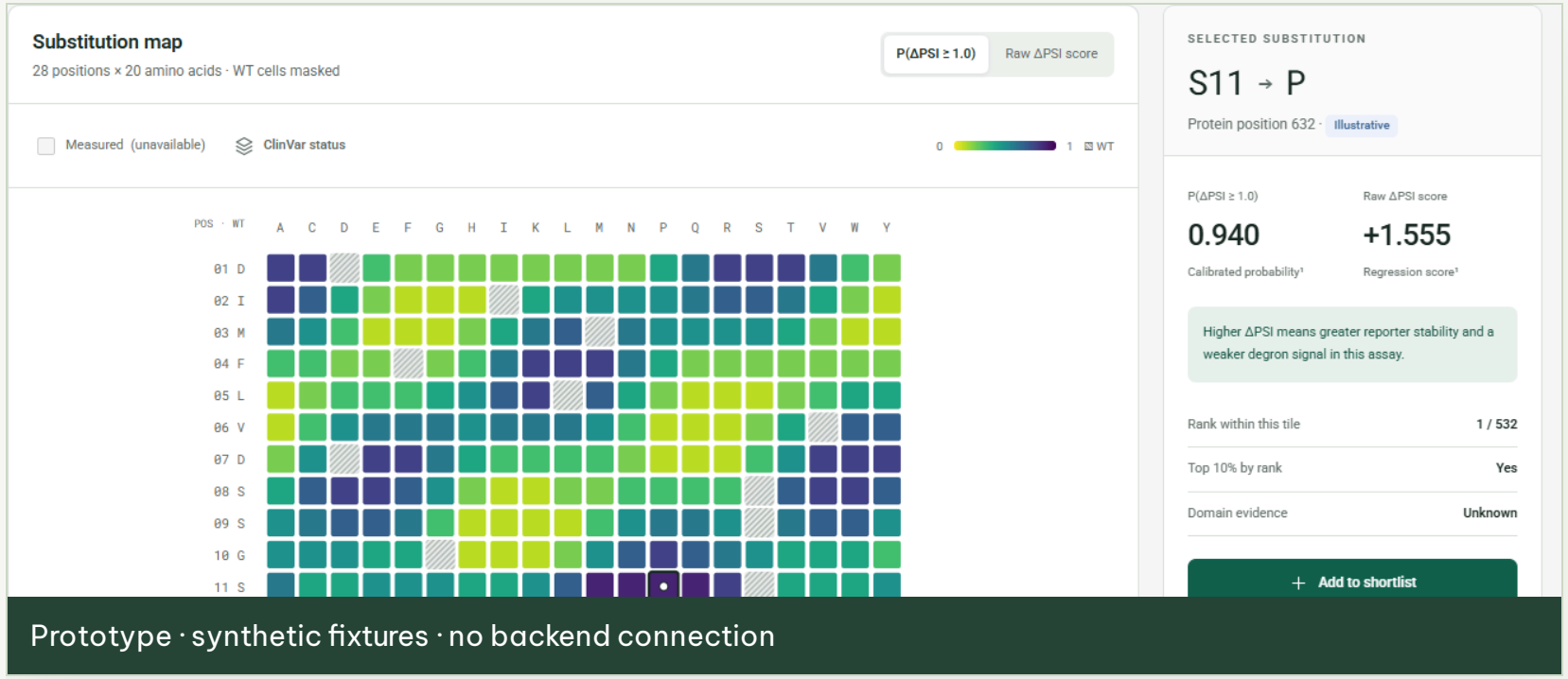
**Select → validate experimentally**

1 · Enter peptide | 2 · View map | 3 · Select variant | 4 · Inspect details | 5 · Export results

Predictions provide a shortlist for follow-up experiments.

ANALYSIS TOOL

# The serving baseline and UI prototype have different roles




Prototype · synthetic fixtures · no backend connection

**Implemented baseline**

React + TypeScript

FastAPI

Serving model:

`m3-ext4.0-platt-v1`

**New research models**

Still require refitting, validation and integration.

Scores in the interface screenshot are synthetic fixtures.

STATUS AND CONTRIBUTIONS

# What is available and what remains unfinished

| Component | Documented status |
| --- | --- |
| **Data, QC and group splits** | Built, with traceability records |
| **Ridge / ESM2 / CNN / mCNN** | Development results available |
| **PLM expansion / frozen–LoRA** | Ongoing; no final conclusion |
| **Calibration / T / external sources** | Incomplete for the new candidate |
| **FastAPI + React baseline** | Implementation available |
| **Workspace and new-model integration** | Implementation / validation pending |

Document snapshot: 3 Nov 2026 · research evidence and software progress are separate.

STATUS AND CONTRIBUTIONS

# Contributions should be grounded in inspectable evidence

01

## Data and benchmarks

Labels, QC, provenance and protein-group splits

02

## Methodological evidence

PSI range, representations, heads and transfer learning

03

## Research tool

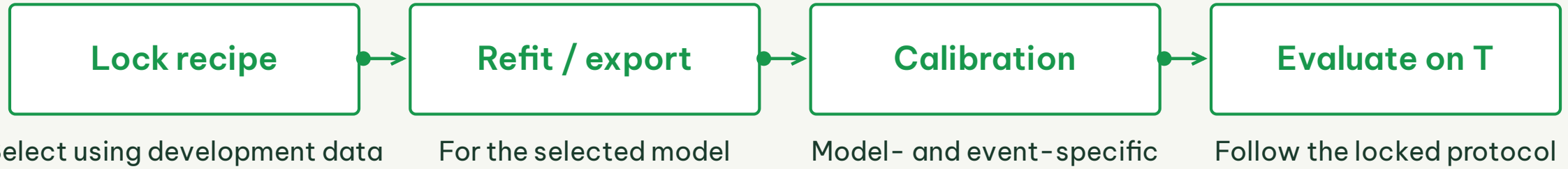
Effect maps, prioritization and traceable results

Research value requires empirical comparisons and clearly bounded conclusions.

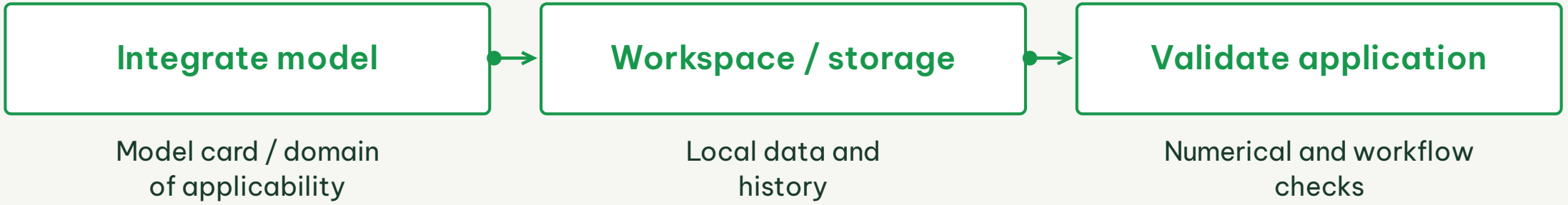
NEXT STEPS

# Lock and validate the model before integration

## MODEL RESEARCH



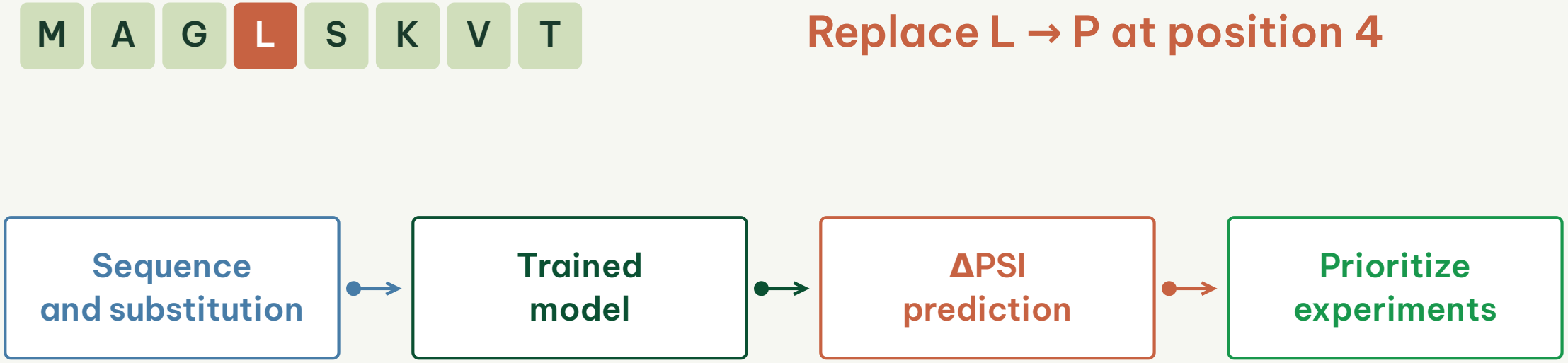

## TOOL DEVELOPMENT




PLM expansion and LoRA need complete comparisons; new models need a serving workflow.

THE COMPLETE RESEARCH QUESTION

# From a sequence to a testable hypothesis




**Available evidence: a development benchmark on Zhang data**

Aim: predict the effects of individual substitutions to support research.

APPENDIX · TERMINOLOGY

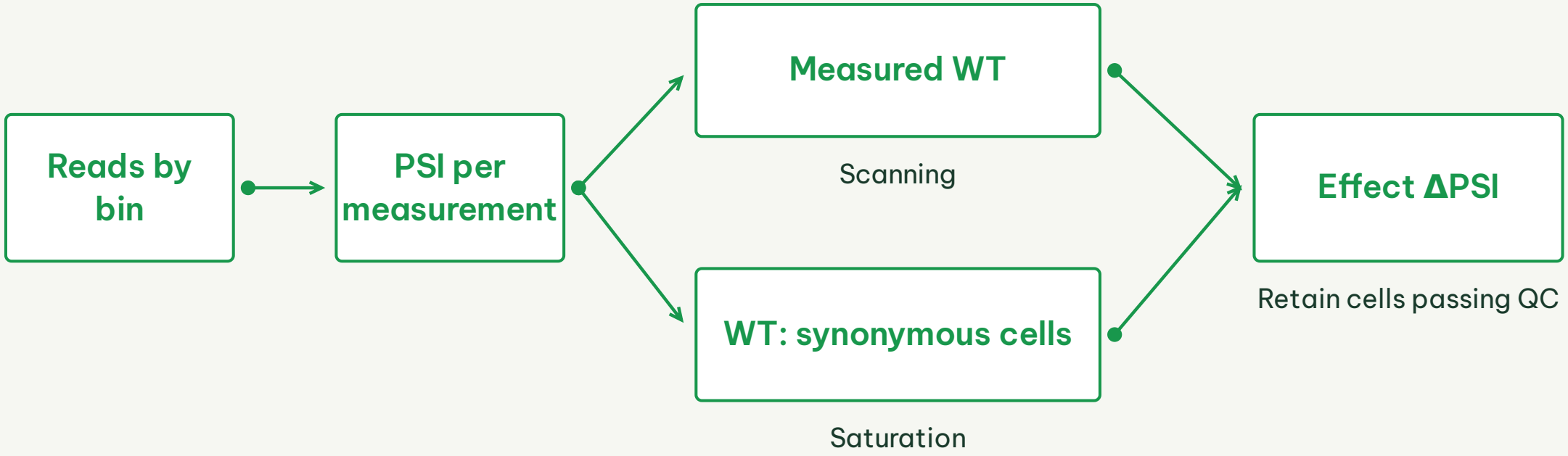
# Key terms used in this presentation

| Term | Meaning in this project |
| --- | --- |
| **Residue / peptide** | One amino acid unit in a chain / a short chain |
| **WT / mutant** | Reference sequence / sequence after substitution |
| **PSI / ΔPSI** | Reporter stability index / change relative to WT |
| **Embedding / head** | Numeric representation / the prediction module using it |
| **Fold / OOF** | A group partition / prediction with the sample held out |
| **Calibration / CI** | Probability calibration / statistical confidence interval |

Reporter stability is the experimental endpoint used here.

APPENDIX · LABEL CONSTRUCTION

# From read counts to ΔPSI: use the correct WT reference

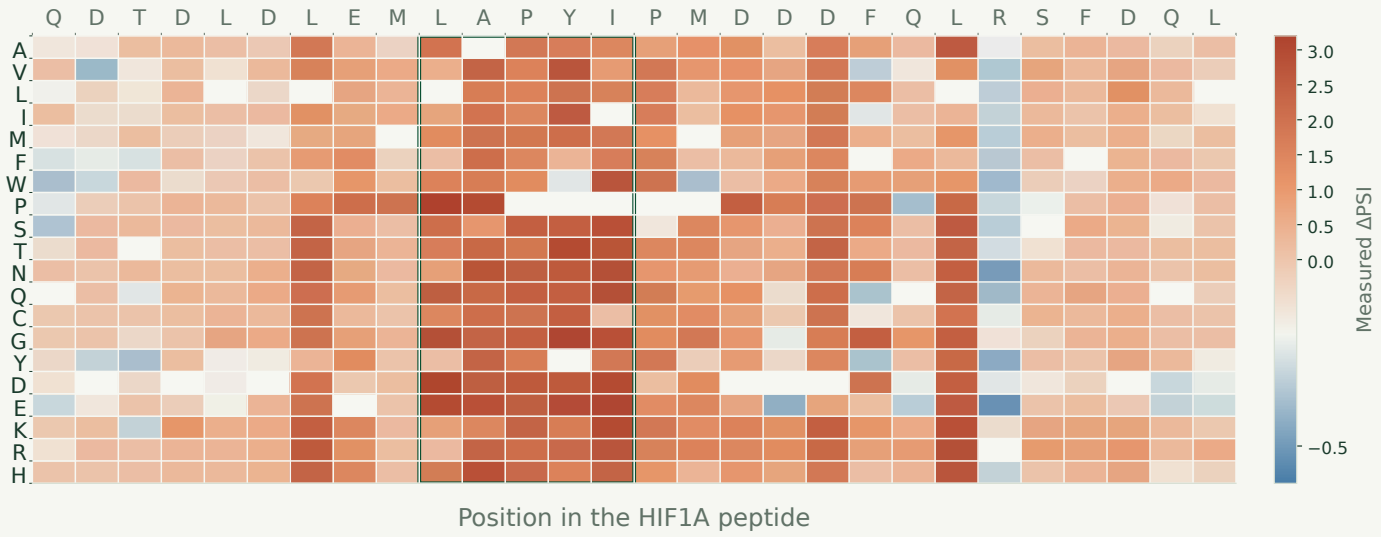



$$PSI = \Sigma(bin \times normalized\ reads) / \Sigma(normalized\ reads)$$

Check read depth, replicates, sequences and coordinates before retaining cells.

APPENDIX · EXPERIMENTAL EXAMPLE

# HIF1A example: reading an experimental effect map




**529 / 532**

substitutions with measured labels

Columns: peptide positions
Rows: replacement AAs
Color: ΔPSI value

Experimental labels

Different positions can respond very differently to substitutions.

APPENDIX · METRICS AND CIS

# Results and units of evaluation

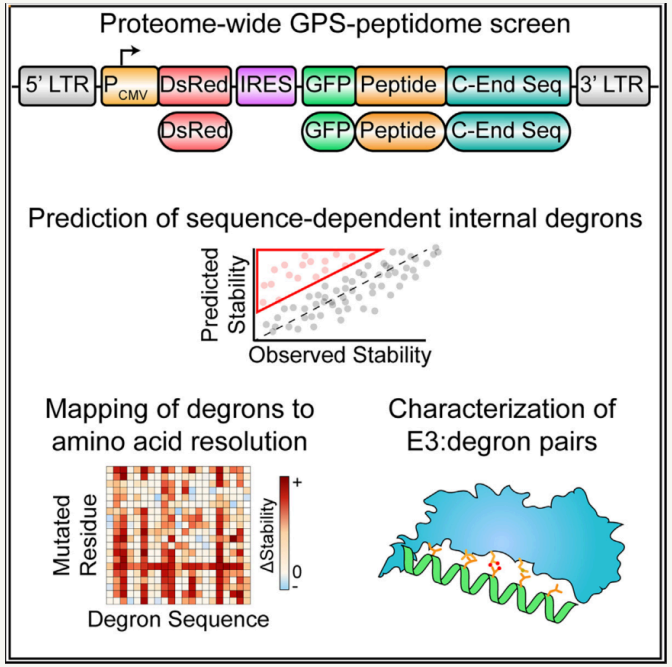
| Model / training scope | MAE ↓ | Positive-event AUROC ↑ | Positive-event AP ↑ |
|---|---|---|---|
| **Context Ridge** | 0.4078 | 0.8319 | 0.4614 |
| **ESM + Pointwise** | 0.3897 | 0.8375 | 0.4739 |
| **ESM + CNN** | 0.3844 | 0.8488 | 0.5095 |
| **ESM + mCNN** | 0.3850 | 0.8463 | 0.5042 |
| **ESM + mCNN, balanced** | 0.3838 | 0.8470 | 0.5064 |

MAE: 235 peptides / 221 groups · positive-event AUROC and AP: 184 eligible peptides
Ensemble seeds 17, 29, 43 · five outer folds · protein-group bootstrap CIs

The first four rows use full-range training; balanced mCNN changes training weights.

APPENDIX · SOURCES

# Sources and evidence can be explored by topic




Zhang et al., 2023 · Graphical abstract · CC BY 4.0

**Original paper and data**
Zhang et al. · Molecular Cell 2023

**Data and QC-filtered counts**
Mutation cells, peptides and protein groups

**Results and comparison conditions**
Kaggle snapshot · analysis dated 2 Nov

**External sources and baseline models**
PAP and Degformer audits; historical records

**Software scope**
Baseline, prototype and outstanding work

**Images, video and asset notices**
Attribution, licenses and asset modifications

Open “Notes” for the current slide's sources and limitations.